# Synthesis

Settings

## Disconnections

Only show ones that fit the selected bond

**FGI**

R–OH → R–X

R–X → R–OH

Alkene → alkane

**OXIDATION**

R–OH → R=O

**REDUCTION**

R=O → R–OH

**FGA**

R–R → R=R

R–R → R–C=O

**SPECIAL REACTIONS**

Diels–Alder

Wittig

**C–X DISCONNECTION**

One group

Two group: 1,1 · 1,2 · 1,3

**C–C DISCONNECTION**

One group

Two group: 1,1 · 1,2 · 1,3

Aldol condensation

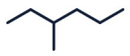
Claisen

More disconnections

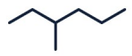
1 Retrosynthesis — 2 Forward synthesis

Restart · New problem

**TARGET**



**SYNTHESIS PLAN**



**INTERMEDIATES (TO DO)**

*None*

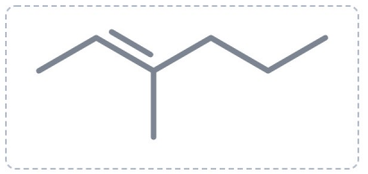
**STARTING MATERIALS**

*None yet*

**TARGET / INTERMEDIATE MOLECULE**



FGA: alkene



Pick a disconnection from the list, then click the bond to apply it to.

Apply disconnection · Clear

Unlimited practice for the organic chemistry problems that trip students up on exams.

**PRACTICE**

IUPAC naming

Synthesis

**LEARN**

Lectures

**ACCOUNT**

Sign in

Settings

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One group

Two group: 1,1 · 1,2 · 1,3

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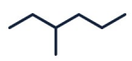
Claisen

More disconnections

1 Retrosynthesis — 2 Forward synthesis

Restart · New problem

**TARGET**



**SYNTHESIS PLAN**




**INTERMEDIATES (TO DO)**

*None*

**STARTING MATERIALS**

*None yet*

**TARGET / INTERMEDIATE MOLECULE**



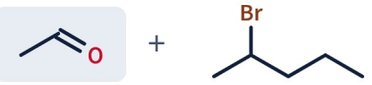
Wittig

**Option A**




OR

**Option B**




Choose which way the bond breaks. Shaded molecules are starting materials, so they don't need to be made.

Use this option · Back

Unlimited practice for the organic chemistry problems that trip students up on exams.

**PRACTICE**
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Lectures

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Sign in
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# Synthesis

Settings

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One group

Two group: 1,1 · 1,2 · 1,3

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One group

Two group: 1,1 · 1,2 · 1,3

Aldol condensation

Claisen

More disconnections

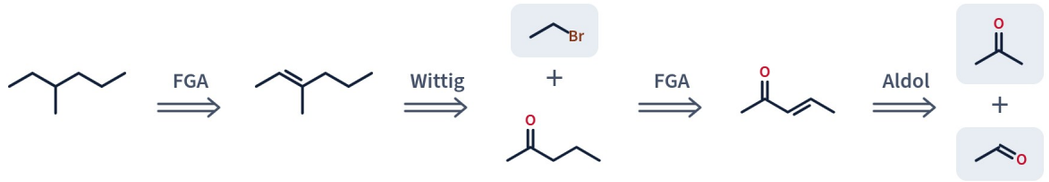
1 Retrosynthesis — 2 Forward synthesis

Restart · New problem

**TARGET**



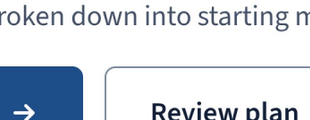
**SYNTHESIS PLAN**



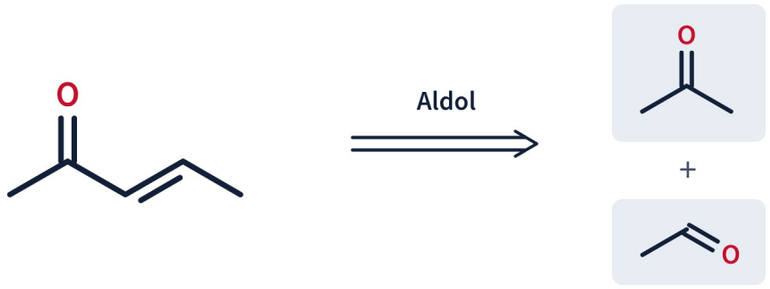

**INTERMEDIATES (TO DO)**

*None*

**STARTING MATERIALS**




**TARGET / INTERMEDIATE MOLECULE**




### Retrosynthesis complete

Next, forward synthesis: match each step with the reagents or conditions that carry it out.

Every intermediate is now broken down into starting materials.

Start forward synthesis → · Review plan



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# Synthesis

Settings

## Reagents and conditions

Search reagents

**CONDITIONS**

Heat (Δ)

Dean–Stark trap (removes water)

$H_3O^+$

NaOH

**REDUCING AGENTS**

$H_2$, Pd/C

$NaBH_4$

$LiAlH_4$

**OXIDIZING AGENTS**

PCC

$KMnO_4$

**PHOSPHORUS REAGENTS**

$PPh_3$, then n-BuLi

**ORGANOMETALLICS**

Mg, ether

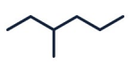
R–Li

Full reagent list

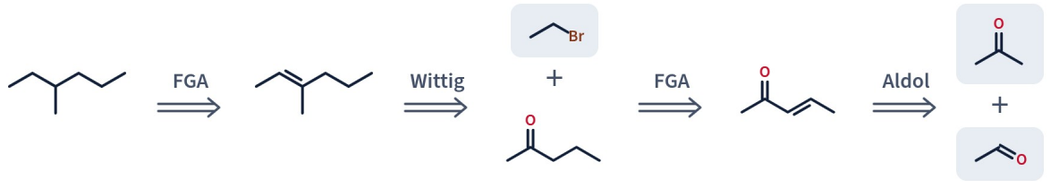
✓ Retrosynthesis — 2 Forward synthesis

Restart · New problem

**TARGET**



**SYNTHESIS PLAN**




**INTERMEDIATES (TO DO)**

*None*

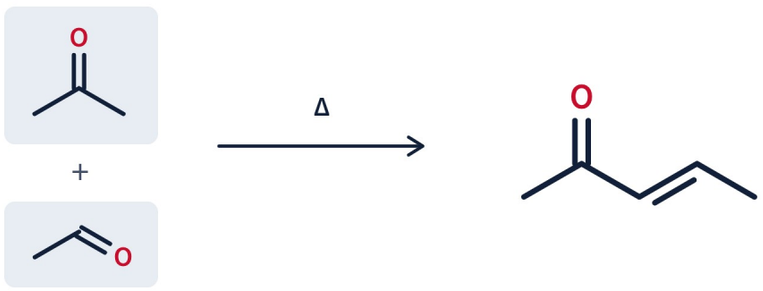
**STARTING MATERIALS**



**FORWARD SYNTHESIS**



**CURRENT STEP**




**Note**

This reaction can give the addition or the elimination product. To favor elimination, use heat (Δ).

No reagent to pick for this step: it uses heat, shown on the arrow.

Next step →



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# Synthesis

Settings

## Reagents and conditions

Search reagents

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$H_3O^+$

NaOH

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$H_2$, Pd/C ×

$NaBH_4$

$LiAlH_4$

**OXIDIZING AGENTS**

PCC

$KMnO_4$

**PHOSPHORUS REAGENTS**

$PPh_3$, then n-BuLi

**ORGANOMETALLICS**

Mg, ether

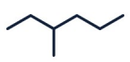
R–Li

Full reagent list

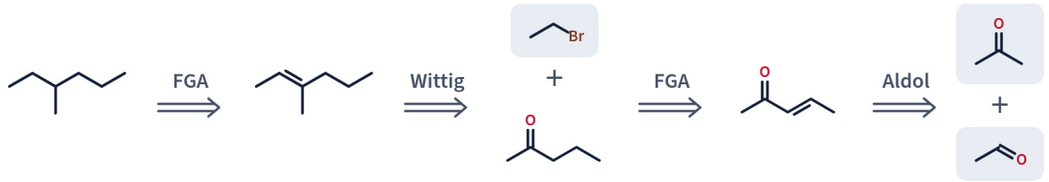
✓ Retrosynthesis — 2 Forward synthesis

Restart · New problem

**TARGET**



**SYNTHESIS PLAN**



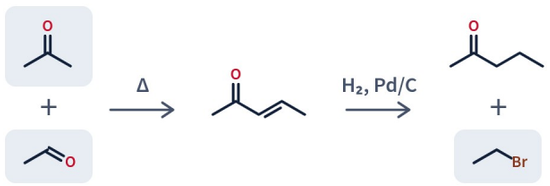

**INTERMEDIATES (TO DO)**

*None*

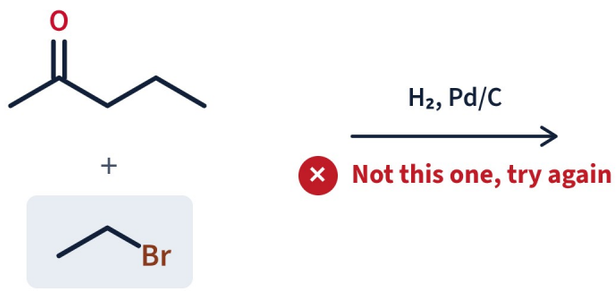
**STARTING MATERIALS**




**FORWARD SYNTHESIS**




**CURRENT STEP**




Pick the reagent or conditions for this step from the list.

Next step →



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# Synthesis

Settings

## Reagents and conditions

Search reagents

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$H_3O^+$

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Mg, ether

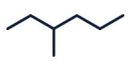
R–Li

Full reagent list

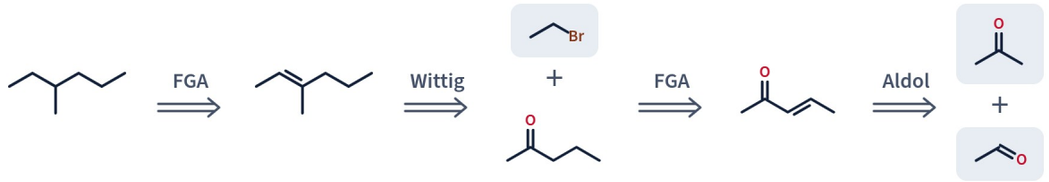
Retrosynthesis — Forward synthesis

Restart  New problem

**TARGET**



**SYNTHESIS PLAN**



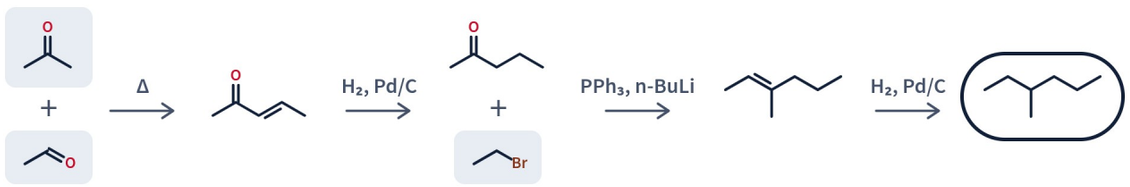

**INTERMEDIATES (TO DO)**

*None*

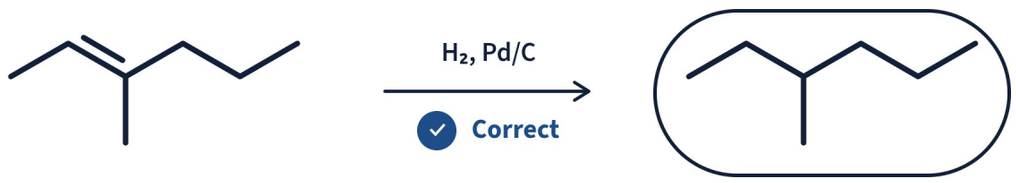
**STARTING MATERIALS**



**FORWARD SYNTHESIS**




**CURRENT STEP**




Correct

## Synthesis complete

You made the target from three starting materials in four steps.

The final product is circled at the end of the forward synthesis row.

New synthesis  Restart this synthesis



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# Synthesis

Step 1 of 2 · **Retrosynthesis**

## SYNTHESIS PLAN

Expand



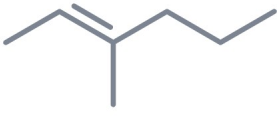
Intermediates: 0 Starting materials: 0

## TARGET MOLECULE





**FGA: alkene**



Tap a bond, then pick a disconnection.

Disconnection

**FGA: R–R → R=R**

Clear

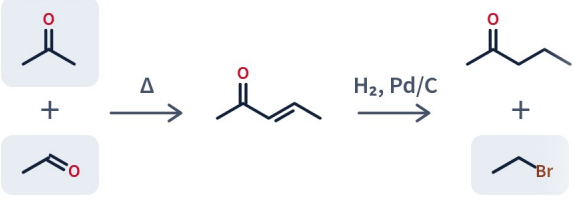
Apply

# Synthesis

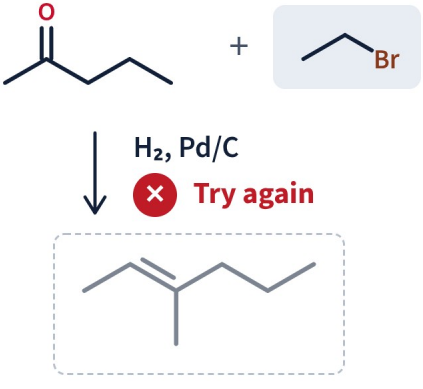
Step 2 of 2 · **Forward synthesis**

## FORWARD SYNTHESIS

Expand




## CURRENT STEP




Reagent

$H_2$, Pd/C

Next step →